

Hypothesis Testing: $x = 1$ vs $y = 0$

$$H_0: x = y \quad | \quad H_a: x \neq y \quad | \quad \alpha = 0.05$$

1. Setup

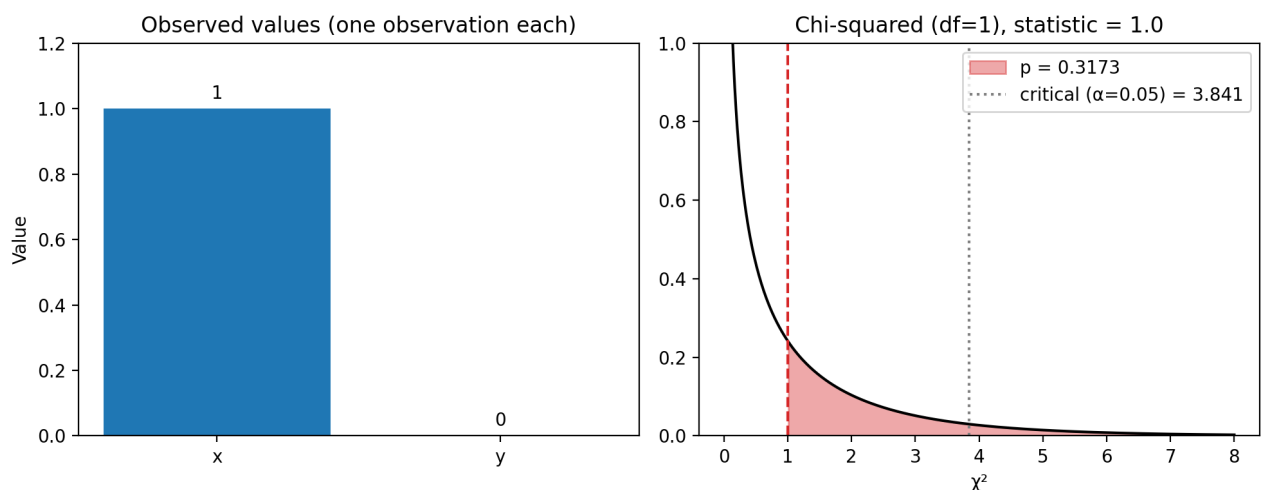
The data are a single observation per variable: $x = 1$ and $y = 0$. The null hypothesis states they are equal; the alternative states they differ. The three tests are applied to exactly this data.

2. Expected results

Test	Statistic	df	p-value	Verdict
Two-sample t-test	undefined (sample variance needs $n \geq 2$ per group)	0	undefined (NaN)	Cannot be run
One-way ANOVA (2 groups)	$SS_{\text{between}} = 0.5$, $SS_{\text{within}} = 0$, F undefined (0 in denominator)	1, 0	undefined (NaN)	Cannot be run
Chi-squared goodness of fit (observed [1, 0], expected [0.5, 0.5])	$\chi^2 = 1.000$	1	0.3173	Fail to reject H_0 , but not valid (expected counts < 5)
Exact binomial test (valid alternative to χ^2)	$k = 1$ of $n = 1$	-	1.0000	Fail to reject H_0

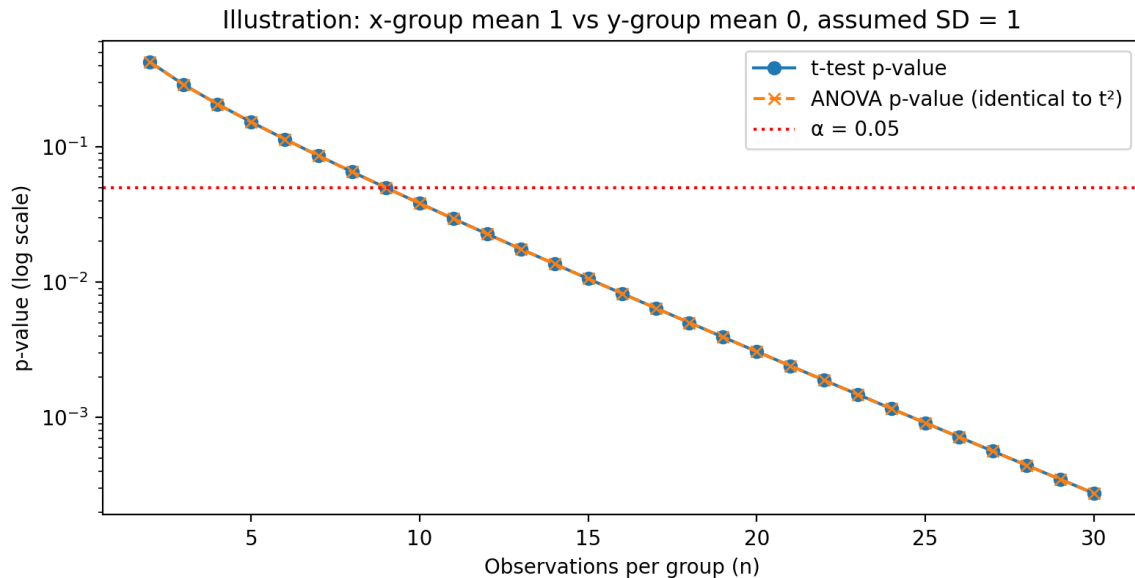
Why t-test and ANOVA fail: both compare the difference between groups with the variability inside groups. With one observation per group, within-group variability is 0 with 0 degrees of freedom, so the test statistic cannot be computed. A difference of 1 cannot be judged without knowing how much values normally fluctuate.

Why chi-squared is only nominal here: it needs counts, so x and y were treated as counts (1 and 0) tested against equal expected counts. The χ^2 approximation requires expected counts of at least about 5; here they are 0.5, so the exact binomial $p = 1.0$ is the trustworthy figure. With a single observation, no test can show a difference.



3. What you would need for the t-test and ANOVA

If x and y were instead *groups* of measurements with means 1 and 0, the p-value would depend on sample size and spread. The graph below is an **illustration with an assumed SD of 1** (not your data): the t-test and ANOVA give the same p-value for two groups ($F = t^2$), and the result first drops below 0.05 at $n = 9$ per group.



4. Conclusion

With $x = 1$ and $y = 0$ as single values, the t-test and ANOVA are undefined, and the chi-squared test ($p \approx 0.317$; exact binomial $p = 1.0$) gives no evidence against H_0 . To test $x \neq y$ properly, collect several observations per group and share them, and I can rerun all three tests on the real data.